

AJ Djalali

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Summary

Research faculty and engineering leader with 12 years building language technology that ships—from formal semantics research at Stanford through two acquisitions (ClearGraph by Tableau, PerceptivePanda by Zapier) and into production systems serving hundreds of thousands of users. PhD in Linguistics, 9 patents in NL interfaces, 15+ peer-reviewed publications. Designed the NLP architecture behind Tableau’s Ask Data before transformer models existed, built deterministic control layers over LLMs grounded in discourse theory, and currently build production AI systems at Georgia Tech—including vector search over 400M images and RAG pipelines on AWS Bedrock. Scaled teams from founding engineer to 30+, led AI strategy across Salesforce’s cloud portfolio, and consult on AI agents and retrieval-augmented generation for startups in maritime and financial services.

Experience

Georgia Institute of Technology — Atlanta, GA

Research Faculty & Senior Research Engineer | Oct 2025 – Present

- Embedded with research teams to build production NLP and AI systems from scratch—not advisory, hands-on engineering
- Designed and built enterprise search & ingestion platform: vector search over 400M+ images via CLIP/Qdrant at sub-200ms latency, distributed GPU ingestion through Ray/Spark on Databricks, zero-error completion across multi-million-image datasets
- Built congressional text analysis platform from the ground up: 181M+ tokens, multi-service architecture with Kafka, Elasticsearch, Neo4j, and spaCy NLP pipelines with full Prometheus/Grafana observability
- Designing graduate curriculum on professional software engineering for AI and distributed systems—15 weeks covering ADRs, trunk-based dev, SRE, and AI-augmented workflows

Gridline — Atlanta, GA

AI Consultant | Oct 2025 – Present

- Built modern RAG system for financial advising on AWS Bedrock with S3-backed vector storage

VesselAware — Remote

Software Engineering Advisor | Jan 2024 – Present

- Guiding architecture and building AI agents for maritime intelligence platform

PerceptivePanda (*acquired by Zapier, Jan 2026*) — San Francisco, CA

Co-Founder & CTO | May 2023 – Jul 2025

- Built AI-native customer research platform replacing human-led interviews with LLM-driven micro-interviews at scale
- Designed the core technical innovation: a deterministic control layer wrapping LLMs using dialogue state management grounded in the Questions Under Discussion model from my Stanford research
- System tracked addressed questions, identified threads for deeper probing, and orchestrated LLM calls through a structured state machine—ensuring analytical coherence while preserving conversational fluidity
- Raised institutional pre-seed funding; accepted into StartX '24 summer cohort

Salesforce — San Francisco, CA

Software Engineering Architect | Jun 2020 – May 2023 *Previously: Principal Member of Technical Staff* | Jun 2020 – Mar 2022

- Led integration of Tableau Ask Data across multiple Salesforce clouds (Sales, Service, Marketing, Commerce)
- Elected to Salesforce AI Northstar Council (2022, 2023)—cross-organization body setting company-wide AI strategy
- Architected next-generation analytics platform spanning CRM Analytics query engine and multiple cloud products
- Led technical integration of Tableau-acquired Narrative Science into the Salesforce ecosystem
- Recognized for Most Patents Filed and Granted across the organization (2022)

Tableau (*via ClearGraph acquisition*) — Seattle, WA / Palo Alto, CA

Staff Software Engineer | Aug 2017 – May 2020

- Created the NLP technology behind Tableau’s Ask Data feature, deployed to hundreds of thousands of users
- Scaled semantic indexing infrastructure using Elasticsearch to support enterprise-scale data catalogs
- Published “Inferencing underspecified natural language utterances in visual analysis” at IUI 2019
- Filed 9 patents covering NLQ architecture, intent inference, cascading edits, table calculations, and incremental visual feedback

ClearGraph (*acquired by Tableau, 2017*) — San Francisco, CA

VP, Natural Language Technology | Nov 2014 – Jul 2017

- Built the core NLP pipeline translating natural language into structured database queries—using Montague Grammar and compositional semantics from my doctoral research, years before BERT existed
- Grew the engineering team from founding stage to 30+ engineers post-acquisition
- Awarded the Celent Model Bank Award (2017) for innovation in financial services technology

Nuance Communications — Burlington, MA

Research Intern, Natural Language Understanding | May – Sep 2014

- Developed NLU prototype based on Synthetic Logic—a formal system I published connecting monotonicity calculi with compositional textual entailment

Education

Stanford University — Stanford, CA

PhD, Linguistics | 2009 – 2015

- Dissertation: *The syntax and semantics of ordinary comparative constructions in English*
- Developed compositional semantic framework using continuation semantics and Montague Grammar
- 5 peer-reviewed publications during doctoral study; Phi Beta Kappa Fellowship (2014)
- Collaborated with Christopher Potts on formal proof systems for natural logic

University of Amsterdam — Amsterdam, NL

Fulbright Scholar — Institute of Logic, Language & Computation | 2008 – 2009

- Research in mathematical logic, formal linguistics, and philosophy of language at the ILLC
- Published on probabilistic inferences in dynamic semantics

Northwestern University — Evanston, IL

BA, Linguistics — Cum Laude with Honors | 2004 – 2008

- Phi Beta Kappa Honor Society; Demoz Paper Award (Linguistics Department)

- Research focus: formal semantics, pragmatics, scalar implicature, bridging inferences

Selected Publications (*15+ total; see Google Scholar*)

- **Djalali, AJ** et al. “Inferencing underspecified natural language utterances in visual analysis.” *Proc. 24th Intl. Conf. on Intelligent User Interfaces (IUI)*, 2019.
 - **Djalali, AJ**. “A constructive solution to the ranking problem in Partial Order Optimality Theory.” *Journal of Logic, Language & Information*, 2017.
 - **Djalali, AJ**. “Synthetic logic.” *Linguistic Issues in Language Technology*, 2014.
 - **Djalali, AJ** & C. Potts. “A Formal Proof System for Natural Logic.” *Workshop on Natural Logic, Proof Theory, and Computational Semantics*, Stanford, 2011.
 - **Djalali, AJ** et al. “Modeling expert effects and common ground using questions under discussion.” *AAAI Workshop on Building Representations of Common Ground with Intelligent Agents*, 2012.
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Patents (*9 issued/pending*)

- 7 granted + 2 pending, all in natural language interfaces for data visualization
 - Covering: NLQ architecture, intent inference, underspecification resolution, cascading edits, table calculations, incremental visual feedback, levels of detail via NL constructs
 - Full listing: alexdjalali.github.io/publications
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Technical Projects

- **House Proceedings Corpus** — Multi-service search platform for 181M+ tokens of congressional text. FastAPI gateway, Kafka-decoupled ingestion, Elasticsearch + Neo4j entity graphs, spaCy NLP pipeline. Full observability with Prometheus/Grafana. github.com/alexdjalali/hpc
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Technical Skills

Domain	Technologies
AI / NLP	LLM orchestration, RAG (AWS Bedrock), AI agents, formal semantics, Montague Grammar, dialogue state management, discourse theory, vector embeddings (CLIP), spaCy, dependency parsing, natural logic, compositional semantics
Languages	Python, Go, TypeScript/JavaScript, Bash, Lua, Java
Systems	FastAPI, React, Next.js, Ray, Spark, Databricks
Data	PostgreSQL, Elasticsearch, Neo4j, Qdrant, Redis, MongoDB, Kafka
Infrastructure	Docker, Kubernetes, Terraform, AWS, Helm, GitHub Actions
Observability	Prometheus, Grafana, OpenTelemetry, structured logging

Honors & Awards

- StartX '24 Summer Cohort
- Salesforce Most Patents Filed & Granted (2022)
- Salesforce AI Northstar Council (2022, 2023)
- Celent Model Bank Award (2017)
- Phi Beta Kappa Fellowship, Stanford (2014)
- Fulbright Fellowship (2008)

- Phi Beta Kappa, Northwestern
- Demoz Paper Award, Northwestern
- National Merit Scholar Semi-Finalist